STUDENTS PERFORMANCE PREDICTION AND PERSONALIZED RECOMMENDATIONS USING MACHINE LEARNING AND FLASK

Submitted to partial fulfillment of the requirements

For the award of a degree of

BACHELOR OF SCIENCE

IN

INFORMATION TECHNOLOGY

Submitted by

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VIVEKANANDHA COLLEG OF ARTS AND SCIENCES FOR WOMEN

(AUTONOMOUS)

[An ISO 9001:2015 Certified Institution]

(Affiliated to Periyar University, Salem, Approved by AICTE,

Re-accredited with "A+" Grade

By NAAC & Recognized U/S 12(B)&2(f) of UGC ACT, 1956)

Elayampalayam, Tiruchengode(Tk), Namakkal (Dt).

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Elayampalayam, Tiruchengode (Tk), Namakkal (Dt).

DEPARTMENT OF COMPUTER SCIENCES AND APPLICATIONS

PROJECT REPORT

CERTIFICATE

The certifies the project work entitled "STUDENTS PERFORMANCE PREDICTION AND PERSONALIZED RECOMMENDATIONS USING MACHINE LEARNING AND FLASK" is a Bonafide record of original work done by DIVYA BHARATHI M S[24UIT04008], RUBA B [24UIT04032], VAISHNAVI M[24UIT04040], during the academic year (2026-2027) partially fulfilled the requirements for the degree of Bachelor of Science in Information Technology.

_____ _____

(Project Guide) (Head of the Department)

This is Submitted for the project viva-voca examination held on _____ at Department of Computer Sciences & Applications, Vivekanandha College of Arts and Sciences for Women (Autonomous), Tiruchengode.

_____ _____

Internal Examiner External Examiner

DECLARATION

I affirm that the project work titled "STUDENTS PERFORMANCE PREDICTION WEBSITE" was submitted to Vivekanandha College of Arts and Sciences For Women (Autonomous) Tiruchengode in partial fulfilment for the award of the degree of Bachelor of Science in Information Technology, is a record of original work done by us under the supervision and guidance of Mr. K. SENTHILKUMAR, MCA., M.Phil.,(Ph.D.) Assistant Professor, Department Of Computer Science & Applications, Vivekanandha College of Arts and Sciences for Women [Autonomous], Tiruchengode. It has not formed the part of any other project work submitted for the award of any degree or diploma, either in this or any other university.

(Signature of the candidate)

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I certify that the declaration made above by the candidate is true to my knowledge.

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ABSTRACT

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Students Performance Prediction and Personalized Recommendations Using Machine Learning and Flask is a web-based system designed to predict students' academic performance and provide personalized suggestions for improvement. In educational institutions, manually monitoring the performance of every student can be difficult and time-consuming. This system uses machine learning techniques to analyse important academic factors such as attendance, internal marks, assignment performance, study hours, previous scores, and subject-wise marks.

The collected student data is processed and provided as input to a trained machine learning model, which predicts the student's expected performance level. Based on the prediction and academic details, the system generates personalized recommendations such as improving attendance, focusing on weaker subjects, completing assignments on time, increasing study hours, and seeking mentor guidance when required.

The application is developed using Python and Flask, providing a simple and interactive web interface for students and mentors. Students can enter their academic details and view their predicted performance and improvement suggestions. Mentors can monitor assigned students and identify those who may require additional academic support.

Overall, the proposed system provides a data-driven and user-friendly approach to student performance monitoring. By combining machine learning with personalized recommendations, it helps students understand their academic strengths and areas for improvement while supporting mentors in providing timely guidance and improving overall learning outcomes.

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INTRODUCTION

CHAPTER 1

INTRODUCTION

1.1 Introduction

The Students Performance Prediction and Personalized Recommendations System is a machine learning-based application developed to predict students' academic performance. It analyses factors such as marks, attendance, and academic activities to identify the expected performance of students.

The system uses Python, Machine Learning, and Flask to provide a simple web-based platform for prediction. It also gives personalized recommendations to help students improve their academic performance and allows mentors to monitor student progress effectively.

1.2 Objectives of the Project

The main objective of this project is to develop an intelligent machine learning-based system for predicting students' academic performance by analysing factors such as marks, attendance, assignments, and other academic activities.

The system aims to identify students who may need additional support and provide personalized recommendations according to their strengths and weaknesses. It also helps mentors monitor student progress, take timely action, and support continuous academic improvement through a simple and user-friendly Flask web application.

1.3 Problem Statement

Traditional methods of evaluating student performance mainly depend on examinations, attendance, assignments, and manual assessment by teachers. These methods may not effectively identify students who are struggling or provide guidance according to each student's individual needs. Therefore, there is a need for an intelligent system that can analyze academic data, predict student performance, identify areas of weakness, and provide personalized recommendations for improvement.

The proposed system uses machine learning for performance prediction and Flask to provide a simple web-based platform for students and mentors to access the results and monitor academic progress.

1.4 Scope of the Project

The scope of this project is to develop a machine learning-based web application that predicts students' academic performance using factors such as marks, attendance, assignments, and other academic details. The system provides personalized recommendations based on individual performance to help students improve their weak areas. Using Flask, the application provides an easy-to-use view predictions, receive recommendations, and monitor academic progress. In the future, the system can be enhanced with larger datasets, advanced machine learning algorithms, and real-time performance tracking.

CHAPTER 2

SYSTEM ANALYSIS

2.1 Existing System

In the existing system, student academic performance is mainly evaluated through examination marks, attendance, assignments, and manual assessments. Teachers and mentors usually analyse these details manually to understand student progress. Some educational systems may use basic statistical methods or separate software to store and analyse student records.

These systems generally have limited personalized analysis and may not provide specific recommendations for each student.

2.2 Disadvantages of Existing System

- **Less personalized**: Individual student needs may not be identified properly.
- **No performance prediction**: Future performance cannot be predicted effectively.
- **Limited recommendations**: Students may not receive suitable improvement suggestions.
- **Difficult monitoring**: Monitoring many students manually is challenging.
- **Human errors**: Manual data handling may lead to errors.
- **Time consuming:** Manual analysis requires more time.

2.3 Proposed System

The proposed system, Student Performance Prediction and Personalized Improvement Recommendation System, is a web-based application designed to predict students' academic performance and provide personalized recommendations for improvement. The system collects student details, academic marks, attendance, study hours, assignment scores, participation in seminars, workshops, quizzes, hackathons, sports, and other extracurricular activities. These details are processed using a Machine Learning model to predict the student's performance level. Based on the predicted results and the student's academic and extracurricular details, the system generates personalized recommendations to help students improve their academic performance and overall development. The application is developed using Python and Flask, with the trained Machine Learning model used for prediction. The proposed system helps students identify areas that need improvement and supports mentors in understanding student performance and providing appropriate guidance. It provides a simple, user-friendly interface

for entering student details, viewing prediction results, and accessing improvement recommendations.

2.4 Advantages of Proposed System

- Accurate performance prediction using machine learning.
- Personalized recommendations based on individual student performance.
- Saves time by reducing manual analysis.
- Easy monitoring of student academic progress.
- Early identification of students who need support.
- User-friendly Flask interface for students and mentors.
- Helps improve academic performance through timely guidance.

CHAPTER 3

SYSTEM REQUIREMENTS

3.1 Hardware Requirements

The hardware requirements for the Students Performance Prediction and Personalized Recommendations System are:

Processor: Intel Core i3 or equivalent

RAM: Minimum 4 GB

Storage: Minimum 10 GB free space

Operating System: Windows 10/11 or equivalent

Display: Standard monitor

Input Devices: Keyboard and mouse

Internet: Required for installing packages and accessing web resources.

3.2 Software Requirements

The software requirements for the Students Performance Prediction and Personalized Recommendations System are:

Operating System: Windows 10/11

Programming Language: Python

Web Framework: Flask

Machine Learning: Scikit-learn

Data Processing: Pandas, NumPy

Model Storage: Joblib

Development Environment: Visual Studio Code

Web Browser: Google Chrome / Microsoft Edge

Database: Excel/CSV files for storing student data.

3.3 Technologies Used

Python: Used for developing the application and implementing machine learning.

Machine Learning: Used to analyse student data and predict academic performance.

Flask: Used to develop the web-based application and connect the machine learning model with the user interface.

Pandas: Used for collecting, processing, and analysing student data.

NumPy: Used for numerical and data operations.

Scikit-learn: Used for training and testing the machine learning model.

HTML & CSS: Used to design the user-friendly web interface.

Joblib: Used to save and load the trained machine learning model.

Excel/CSV: Used for storing and managing student academic data.

3.4 About Python

Python is a high-level, interpreted, and easy-to-learn programming language widely used in software development, data analysis, and machine learning.

In this project, Python is used to process student data, train the machine learning model, predict academic performance, and generate personalized recommendations. Python libraries such as Pandas, NumPy, and Scikit-learn are used for data processing and machine learning, while Flask is used to create the web application.

3.5 About Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and make predictions or decisions without being explicitly programmed for every task.

In this project, machine learning is used to analyse students' academic data and predict their performance. The trained model uses factors such as marks, attendance, and assignments to generate predictions. Based on the prediction, the system provides personalized recommendations to help students improve their academic performance.

3.6 About Flask

Flask is a lightweight and flexible web framework for Python used to develop web applications. It provides tools for creating web pages, handling user requests, and connecting applications with databases or machine learning models.

In this project, Flask is used to create the web interface, accept student academic details, connect the trained machine learning model, display performance predictions, and provide personalized recommendations. It makes the system simple and easy for students and mentors to use.

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

The system architecture describes the flow of the Students Performance Prediction and Personalized Recommendations System. The system collects student academic details such as marks, attendance, and assignments through the Flask web interface. The data is processed and given to the trained machine learning model for performance prediction. Based on the prediction and identified weak areas, the system generates personalized recommendations. The final prediction and recommendations are displayed to the student or mentor.

4.2 Data Flow Diagram

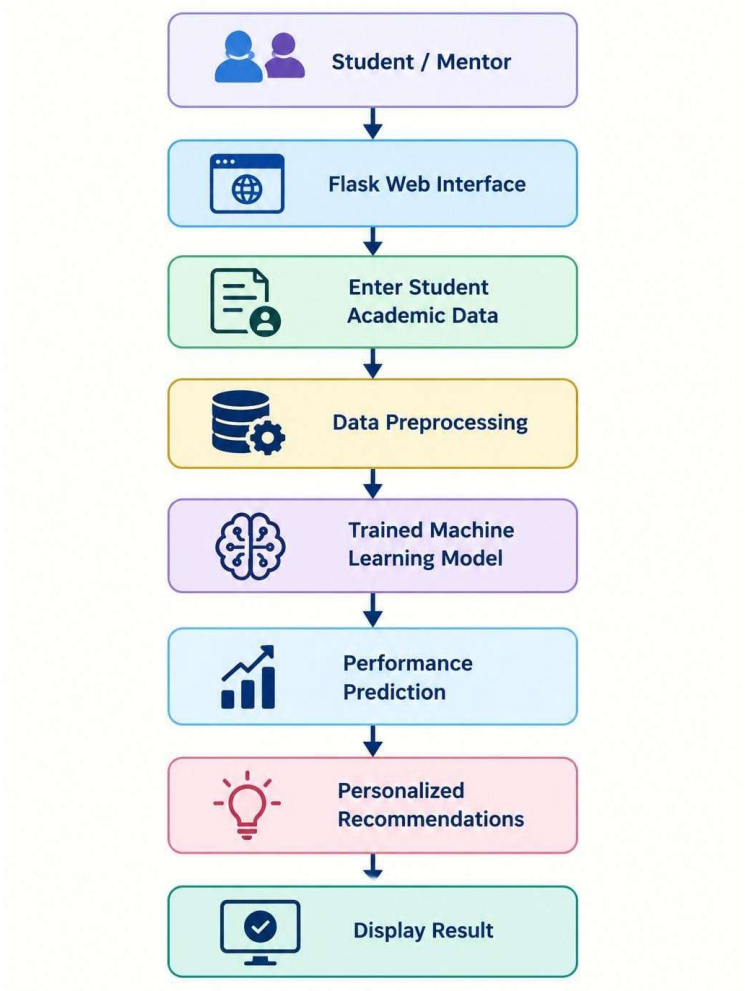


Figure 4.2.1: This diagram shows the flow of student data, performance prediction, and personalized recommendations.

4.3 USE CASE DIAGRAM:

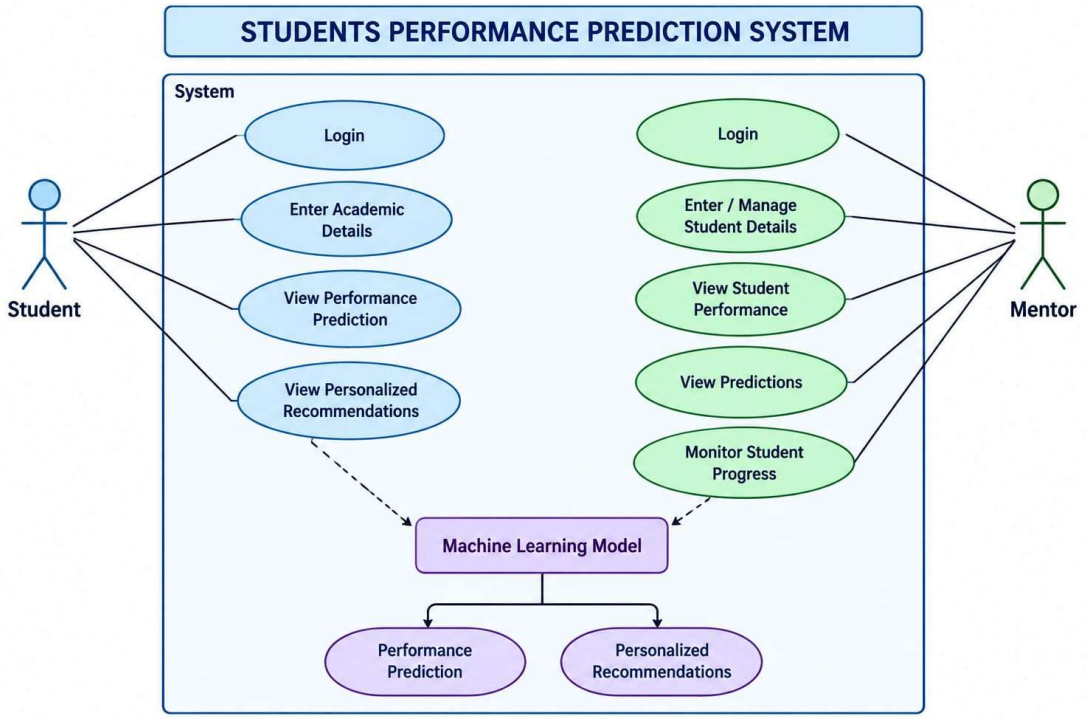


Figure 4.3.1: This diagram shows the roles of students and mentors in predicting performance and providing personalized recommendations.

TABLE 4.1 Prediction & Recommendations Table:

Field Name	Data Type	Description
Prediction_ID	Integer	Unique Prediction ID
Student_ID	Integer	Student Identification
Predicted_ Performance	String	Predicted Performance Level
Weak_area	String	Identified area for Improve
Recommendations	String	Personalized improvement suggestion

Table 4.2 Database Design

FIELD NAME	DATA TYPE	DESCRIPTION
Student_Id	Integer	Unique Student identification
Student_name	String	Name of the student
Attendence	Float	Attendance percentage
Assignment_Marks	Float	Assignment performance
Internal_Marks	Float	Internal Examination Marks
Subject_Marks	Float	Subject Wise Marks
Overall_Marks	Float	Overall Academic Marks

4.5 Module Description

The Students Performance Prediction and Personalized Recommendations System consists of the following modules:

1. Student Data Collection Module

This module collects student details such as marks, attendance, assignment scores, and

2. Data Preprocessing Module

The collected data is cleaned and prepared before giving it to the machine learning model.

3. Performance Prediction Module

Machine learning model analyses the student data and the predicts the expected academic performance.

4. Personalized Recommendation Module

Based on the prediction and performance areas, the system provides personalized suggestions to help students improve.

5. Mentor Monitoring Module

This module allows mentors to view student details, predictions, recommendations, and academic progress.

6. Flask Web Application Module

Flask connects all the modules and provides a simple web interface through which students and mentors can access the system.

CHAPTER 5

SYSTEM IMPLEMENTATIONS

5.1 Data Collection and Preprocessing

The system collects student academic data such as marks, attendance, assignment scores, and other academic details. The collected data is organized in a structured format using Excel or CSV files.

The data is then pre-processed by removing missing or incorrect values, converting data into a suitable format, and selecting the important features required for prediction. The processed data is used to train the machine learning model and generate accurate performance predictions and personalized recommendations.

5.2 Model Training and Prediction

The pre-processed student data is used to train a machine learning model to identify patterns in academic performance. The trained model learns from factors such as marks, attendance, and assignment scores. After training, new student data is given to the model to predict the student's expected performance. The prediction is then displayed through the Flask web application and used to provide suitable personalized recommendations.

5.3 Personalized Recommendation Generations

The system generates personalized recommendations based on each student's predicted performance and identified weak areas. It analyses factors such as marks, attendance, and assignment performance to provide suitable suggestions. These recommendations help students focus on their weak subjects, improve their study habits, and achieve better academic performance. The recommendations are displayed through the Flask web application along with the prediction results.

5.4 Flask Web Applications

The Flask web application provides a simple and user-friendly interface for the Students Performance Prediction system. Users can enter student academic details such as marks, attendance, and assignment scores through the web interface. Flask connects the entered data with the trained machine learning model and displays the predicted performance and personalized recommendations. It also helps mentors view student details and monitor their academic progress efficiently.

5.5 Student and Mentor Dashboards

The system provides separate dashboards for students and mentors. The Student Dashboard allows students to enter their academic details, view their predicted performance, and receive personalized recommendations for improvement. The Mentor Dashboard allows mentors to view assigned students, check their academic performance and predictions, and monitor their progress. These dashboards make the system easy to use and support effective academic monitoring and guidance.

Dashboard Architecture Diagram

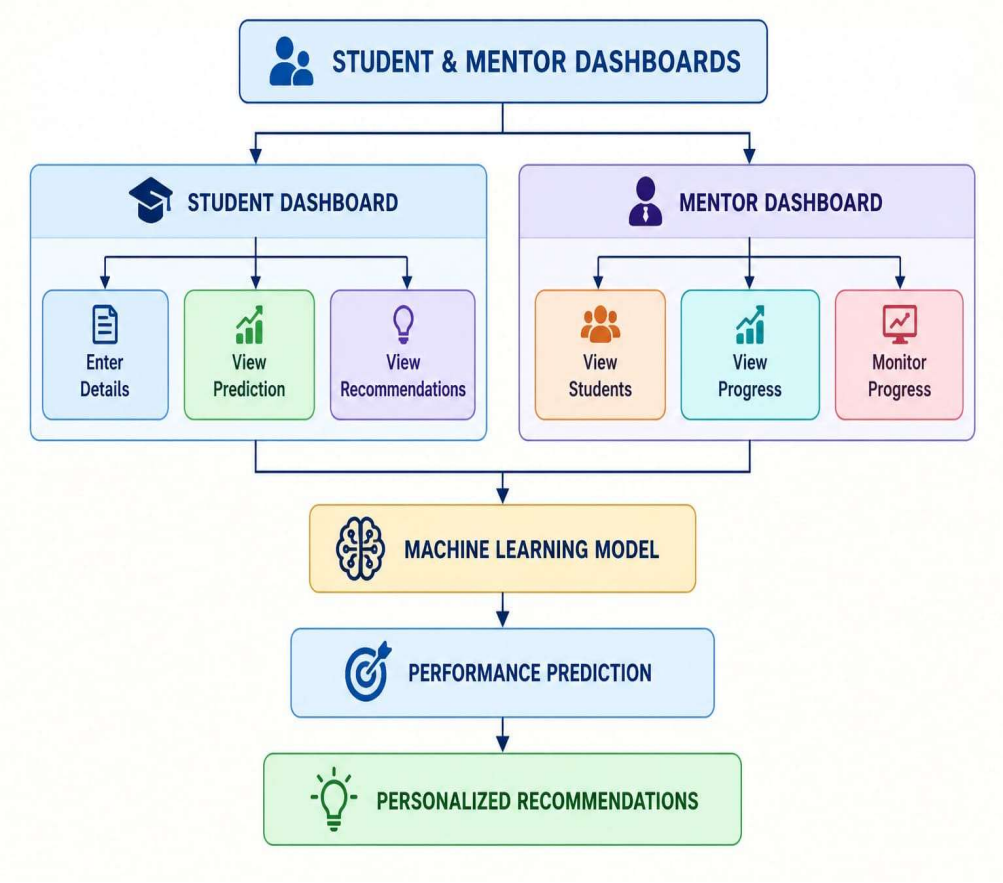


Figure 5.5.1: This diagram shows the student and mentor dashboards, performance prediction, and personalized recommendations.

CHAPTER 6

TESTING AND RESULT

6.1 Testing Methodology

Testing is performed to ensure that the system works correctly and gives expected results. Different student academic details are entered and the prediction and recommendations are checked. The login, dashboards, data processing, prediction, and recommendation modules are also tested to identify and correct errors.

6.2 Test Cases

The following test cases are used to verify the functionality of the Students Performance Prediction and Personalized Recommendations Using Machine Learning and Flask system.

Test Case	Test Description	Expected Result	Status
TC01	Student login with valid details	Student dashboard should open successfully	Pass
TC02	Mentor login with valid details	Mentor dashboards open successfully	Pass
TC03	Enter valid student academic details	Details should be accepted successfully	Pass
TC04	Enter incomplete/invalid details	System should display on appropriate error messages	Pass
TC05	Predict Student Performance	System should display the predicted performance	Pass
TC06	Generate Personalized Recommendations	Suitable Improvement recommendations should be displayed	Pass
TC07	View assigned students in mentor dashboard	Assigned student details should be displayed	Pass
TC08	Monitor student Performance	Mentor should be able to view student performance details	Pass
TC09	Logout from the system	User should be logged out successfully	Pass

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

The Students Performance Prediction and Personalized Recommendations Using Machine Learning and Flask system provide an effective way to analyse and predict students' academic performance. The system uses academic data such as marks, attendance, and assignment scores to generate performance predictions and personalized recommendations. The Flask web application makes the system simple and user-friendly for students and mentors. Overall, the system helps identify areas for improvement and supports better academic monitoring and guidance.

7.2 Future Enhancements

In the future, the system can be enhanced by implementing advanced machine learning algorithms to improve prediction accuracy. Mobile application integration, real-time performance monitoring, automated alerts, and personalized learning recommendations can be introduced. Integration with college management systems can also simplify data collection and help students improve their academic performance effectively.

SOURCE CODE

```
from flask import Flask, request, redirect, url_for, session, render_template_string
import joblib
import pandas as pd
import os
import glob
import re

# Flask Application
app = Flask(__name__)
app.secret_key = "student_performance_system_2026"

# Load trained model and feature columns
BASE_DIR = os.path.dirname(os.path.abspath(__file__))

MODEL_PATH = os.path.join(BASE_DIR, "student_performance_model.pkl")
FEATURE_PATH = os.path.join(BASE_DIR, "feature_columns.pkl")

model = joblib.load(MODEL_PATH)
feature_columns = joblib.load(FEATURE_PATH)

# Load Student Dataset
excel_files = []

for ext in ["*.xlsx", "*.xlsm", "*.xls"]:
    excel_files.extend(
        glob.glob(os.path.join(BASE_DIR, ext))
    )

if not excel_files:
    raise FileNotFoundError("Excel dataset not found.")

DATASET_PATH = excel_files[0]
```

```
df = pd.read_excel(DATASET_PATH)

# Find Student Columns
def find_column(possible_names):
    for name in possible_names:
        if name in df.columns:
            return name

    lower_map = {
        str(c).strip().lower(): c
        for c in df.columns
    }

    for name in possible_names:
        if name.lower() in lower_map:
            return lower_map[name.lower()]

    return None

REGISTER_COL = find_column([
    "Student ID",
    "Register No",
    "Register Number",
    "Registration Number"
])

NAME_COL = find_column([
    "Student Name",
    "Name"
])

# Clean Register Number
```

```
def clean_register(value):
    if pd.isna(value):
        return ""

    value = str(value).strip()

    if value.endswith(".0"):
        value = value[:-2]

    return value

df[REGISTER_COL] = df[REGISTER_COL].apply(clean_register)

# Find Student
def find_student(register_no):
    register_no = clean_register(register_no)

    result = df[
        df[REGISTER_COL].astype(str) == register_no
    ]

    if result.empty:
        return None

    return result.iloc[0]

# Performance Classification
def get_performance(mark):

    try:
        mark = float(mark)
    except:
```

```
        mark = 0

    if mark >= 75:
        return "Excellent"
    elif mark >= 60:
        return "Good"
    elif mark >= 50:
        return "Average"
    else:
        return "Need Improvement"

# Personalized Recommendations
def generate_recommendations(academic, activities):

    recommendations = []

    attendance = float(
        academic.get("attendance", 0)
    )

    study_hours = float(
        academic.get("study_hours", 0)
    )

    assignment_score = float(
        academic.get("assignment_score", 0)
    )

    programming = float(
        academic.get("programming", 0)
    )
```

```
data_science = float(
    academic.get("data_science", 0)
)

previous_mark = float(
    academic.get("previous_mark", 0)
)

if attendance < 75:
    recommendations.append(
        "Improve regular attendance."
    )
else:
    recommendations.append(
        "Maintain good attendance."
    )

if study_hours < 3:
    recommendations.append(
        "Increase your regular study schedule."
    )

if assignment_score < 60:
    recommendations.append(
        "Complete assignments accurately and on time."
    )

if programming < 60:
    recommendations.append(
        "Practice programming problems regularly."
    )
```

```
    if data_science < 60:
        recommendations.append(
            "Practice Data Science concepts."
        )

    if previous_mark < 60:
        recommendations.append(
            "Improve previous semester performance."
        )

    return recommendations[:6]

# Student Performance Prediction
def predict_performance(student_data):

    input_data = pd.DataFrame(
        [student_data]
    )

    input_data = input_data[
        feature_columns
    ]

    prediction = model.predict(input_data)

    return prediction[0]

# Flask Home Page
@app.route("/")
def home():
    return render_template_string(
        """
```

```
        <h1>Students Performance Prediction</h1>
        <p>Machine Learning Based System</p>
        """
    )

# Prediction Route
@app.route("/predict", methods=["POST"])
def predict():

    student_data = {}

    for feature in feature_columns:
        value = request.form.get(
            feature, 0
        )

        try:
            value = float(value)
        except:
            value = 0

        student_data[feature] = value

    prediction = predict_performance(
        student_data
    )

    performance = get_performance(
        prediction
    )

    recommendations = generate_recommendations(
```

```
        student_data,
        {}
    )

    return render_template_string(
        """
        <h2>Performance Prediction</h2>

        <p>
        Predicted Performance:
        <b>{{ performance }}</b>
        </p>

        <h3>Personalized Recommendations</h3>

        <ul>
        {% for recommendation in recommendations %}
            <li>{{ recommendation }}</li>
        {% endfor %}
        </ul>
        """,
        performance=performance,
        recommendations=recommendations
    )

# Run Application
if __name__ == "__main__":
    app.run(
        debug=True
    )
```

SAMPLE OUTPUT

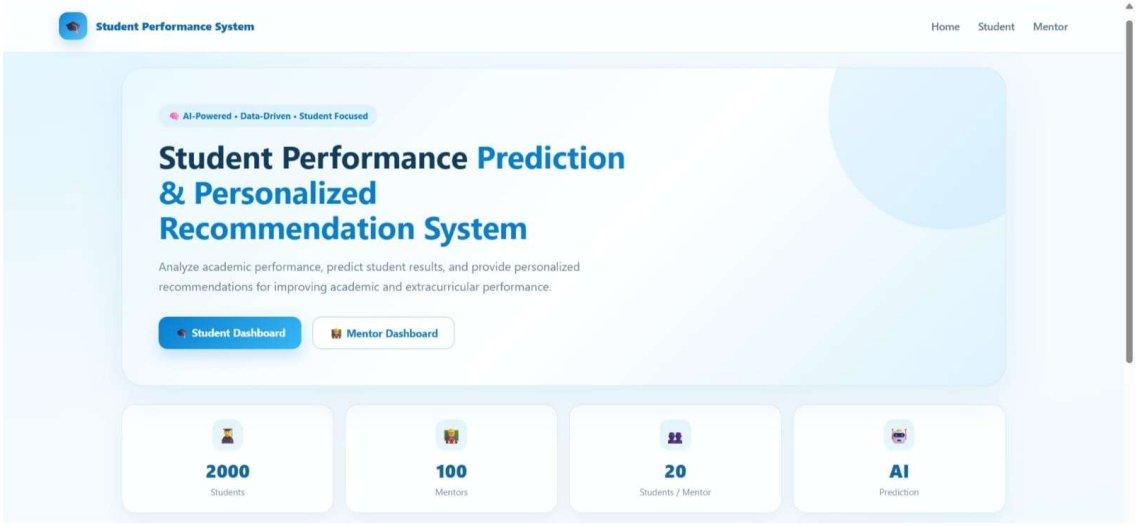


Figure 1.1 :Home page:Displays the system introduction and dashboard options.

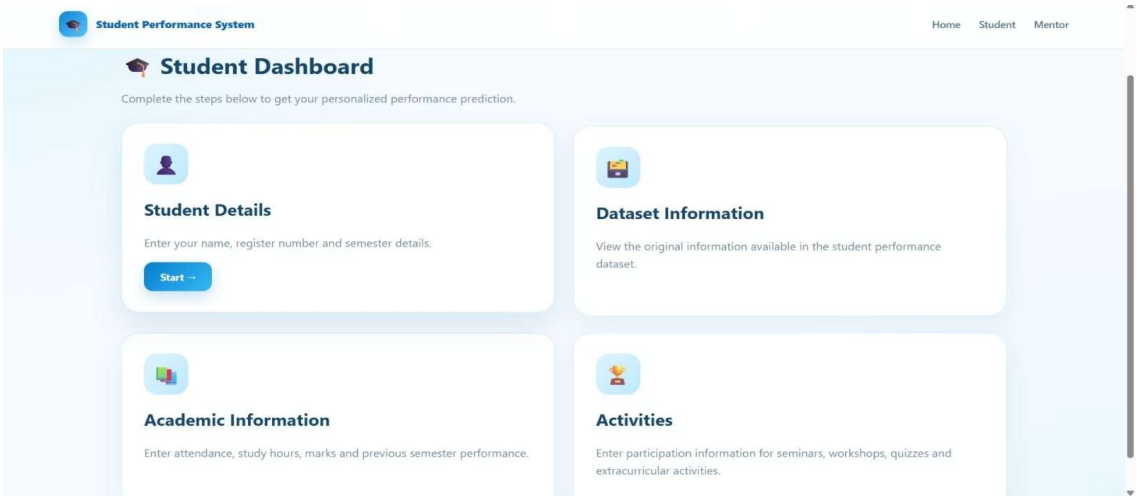


Figure 1.2: Student Dashboard shows the main modules of the system.

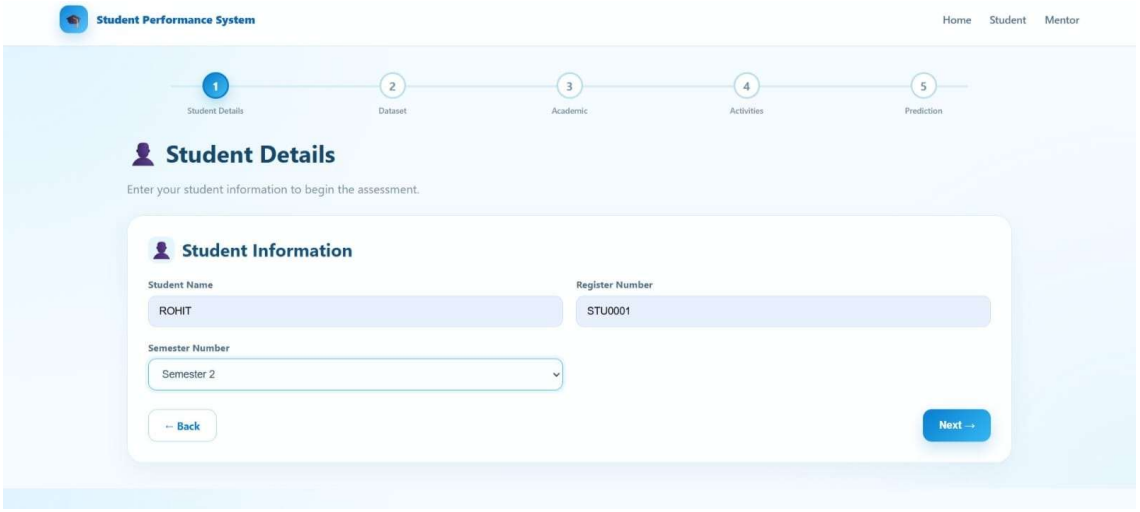


Figure 1.3: Student Details: collects the student's name, register number, and semester number.

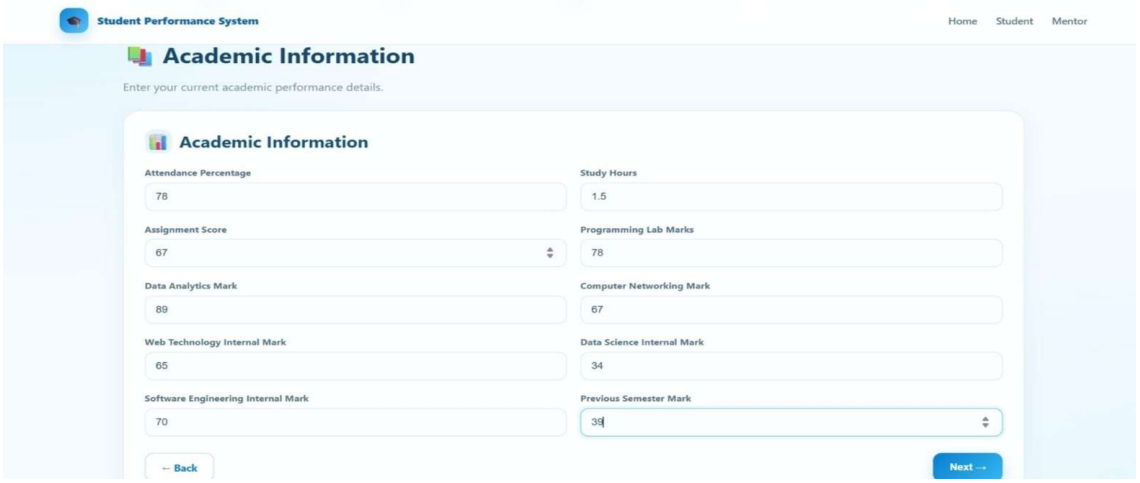


Figure 1.4: Academic information: Displays the student's attendance, study hours, and subject marks.

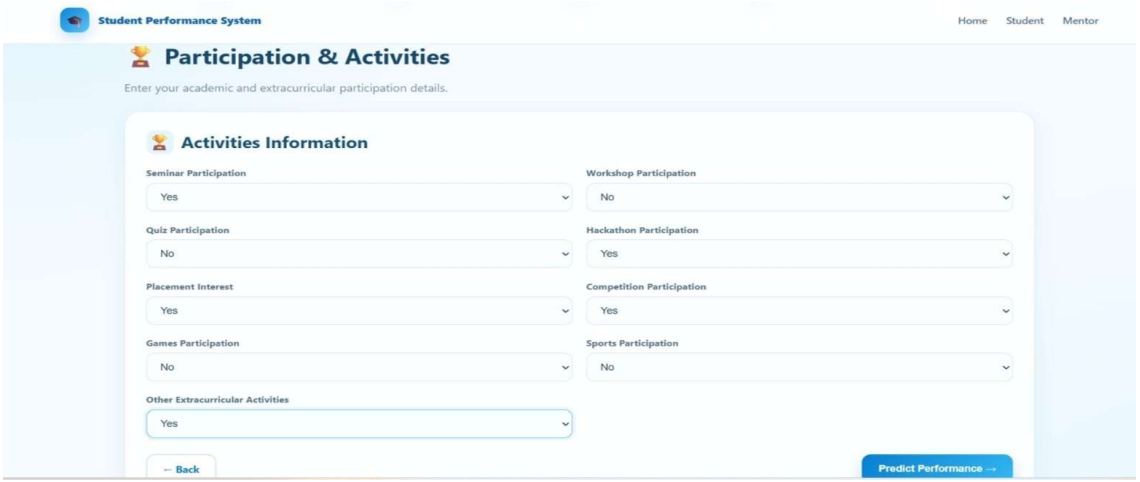


Figure 1.5: Participation & Activities: collects the student's participation in extracurricular activities.

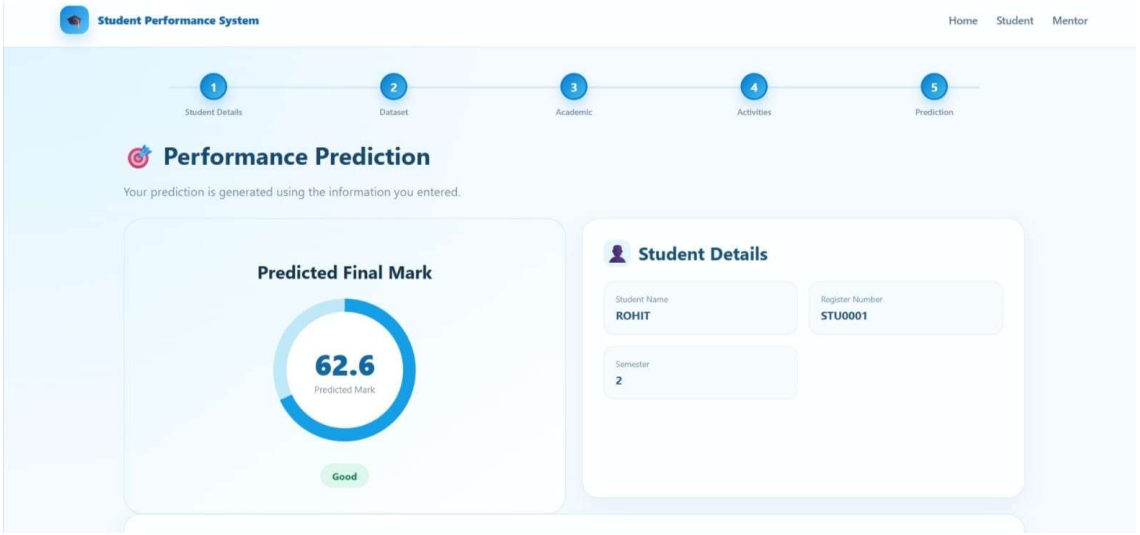


Figure 1.6: performance prediction: Display the student's predicted final mark and performance level.

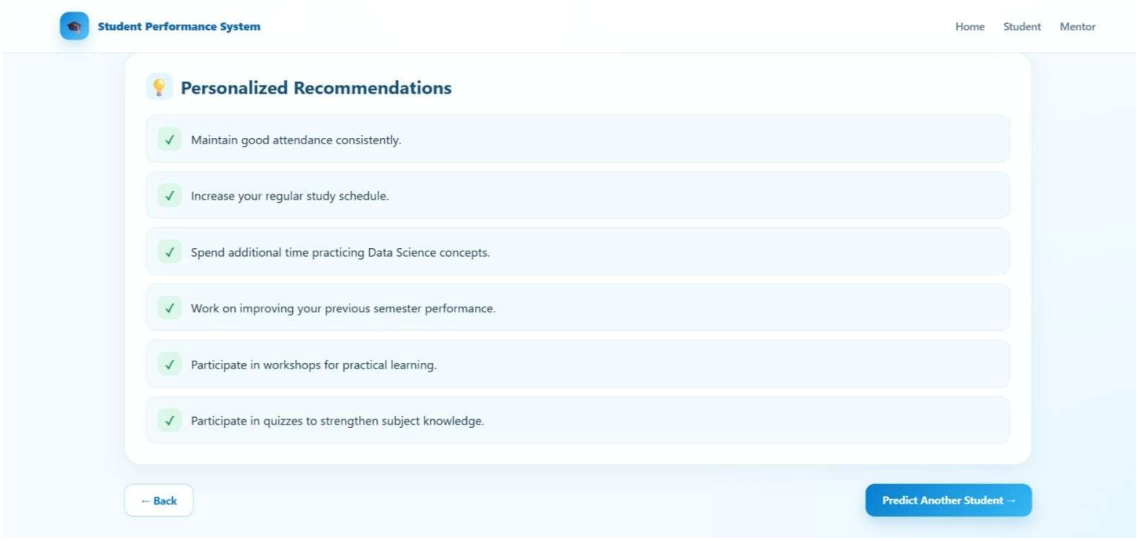


Figure 1.7: Personalized Recommendations: provides suggestion to improve the student's academic performance and extracurricular participation.

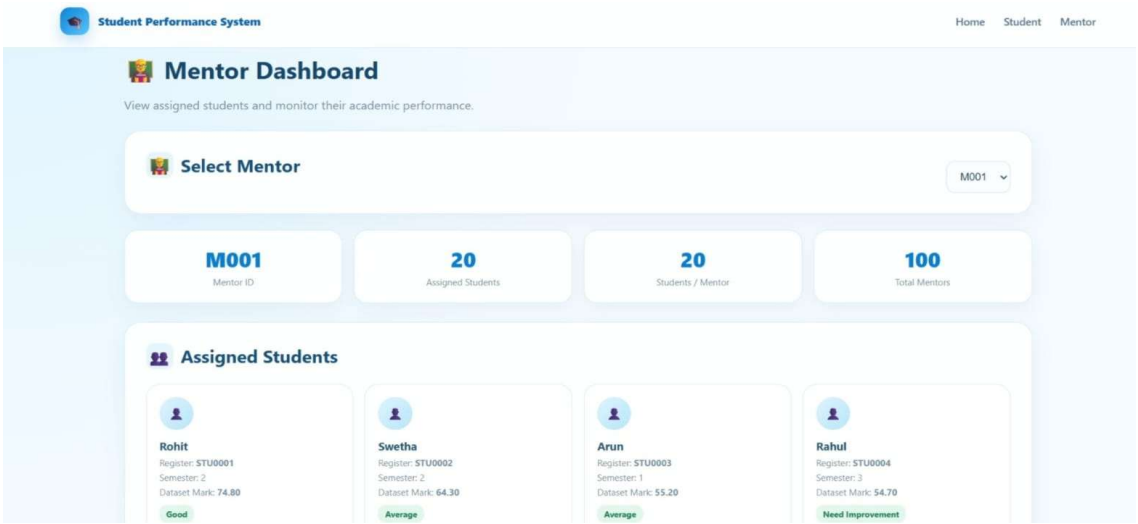


Figure 1.8: Mentor Dashboard: Displays the mentor's details and assigned students with their academic performance.

REFERENCES

BOOK REFERENCES

1. Géron, A. (2022). Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media.

2. Peña-Ayala, A. (Ed.). (2014). Educational Data Mining: Applications and Trends. Springer.

3. Ricci, F., Rokach, L., & Shapira, B. (Eds.). (2022). Recommender Systems Handbook. Springer.

4. McKinney, W. (2022). Python for Data Analysis. O'Reilly Media.

5. Han, J., Kamber, M., & Pei, J. (2011). Data Mining: Concepts and Techniques. Morgan Kaufmann.

6. Aggarwal, C. C. (2015). Data Mining: The Textbook. Springer.

7. James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning. Springer.

8. Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Springer.

9. Grus, J. (2019). Data Science from Scratch. O'Reilly Media.

10. Lutz, M. (2013). Learning Python. O'Reilly Media.

11. Grinberg, M. (2018). Flask Web Development: Developing Web Applications with Python. O'Reilly Media.

12. Raschka, S., Liu, Y., & Mir Jalili, V. (2022). Machine Learning with PyTorch and Scikit-Learn. Packt Publishing.

13. Provost, F., & Fawcett, T. (2013). Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking. O'Reilly Media.

14. Kelleher, J. D., Namee, B. M., & D'Arcy, A. (2020). Fundamentals of Machine Learning for Predictive Data Analytics. MIT Press.

WEB REFERENCES

1. Python Software Foundation, "Python Documentation." Available at: https://docs.python.org/3/

2. Pallets Projects, “Flask Documentation.” Available at: https://flask.palletsprojects.com/

3. Scikit-learn Developers, “Scikit-learn User Guide.” Available at: https://scikit-learn.org/stable/user_guide.html

4. The pandas Development Team, “Pandas Documentation.” Available at: https://pandas.pydata.org/docs/

5. NumPy Developers, “NumPy Documentation.” Available at: https://numpy.org/doc/

6. Joblib Developers, “Joblib Documentation.” Available at: https://joblib.readthedocs.io/

7. Mozilla Developer Network, “HTML Documentation.” Available at: https://developer.mozilla.org/en-US/docs/Web/HTML

8. Mozilla Developer Network, “CSS Documentation.” Available at: https://developer.mozilla.org/en-US/docs/Web/CSS

9. GitHub, “GitHub Documentation.” Available at: https://docs.github.com/